

Element E: Application of STEM Principles and Practices

Our design integrates various engineering and mathematical principles. We calculated optimal plant sizes using spatial efficiency equations to maximize the use of available space. For plant selection, we researched the height, amount of space, and the amount of water needed for each plant, and the time they take to grow. Additionally, we incorporated height restrictions and space efficiency elements to ensure beauty and meet the requirements for the plant beds. Through different examples of water beds, we refined our plants and spacing and tested different configurations before finalizing our prototype.